Canonical oriented state without equation history

T-1869 · 2026-11-07 · Prime-field compressed recovery

Replace a growing list of pair equations with disjoint monic supports B (both roots) and S (one selected root), plus the selected root function v modulo S. Ramified fibers W=±2 are selected from initialization, making the representation independent of update order.

Canonical state: both-root B(W), selected-root S(W), root v(W)

Initialize ramified W=±2 as selected; cache G=B·S for mode evaluation
No retained equation list; each update consumes a(W)+b(W)V

Selected: substitute v(W). Both: split b=0 / b≠0, then test norm

New selected roots −a/b ⟶ coprime CRT merge with old selected roots

Update B,S,v,G ⟶ later radius or capped oriented lift

2,516exact states · history-slot sum75,387 ⟶ canonical58,119 (not peak bytes)

Exact update. Selected roots are filtered by gcd(S,a+bv). On B, split where b vanishes: retain both roots only if a also vanishes. Else select −a/b where its quadratic norm is zero. Invert only after restricting to a unit component; merge old and new selected roots by coprime CRT. This preserves exactly the same candidate set and removes all prior-equation storage.

Native exact control	Result
Candidate-membership checks	852
Reverse-history / duplicate controls	3 / 18
Additional zero equations without history growth	96
States matching persistent reference	2,516
Live coefficient-slot sums: reference / canonical	75,387 / 58,119

Public recovery. Both281-target panels retain all scalars and routes: cap64uses2,881source calls; cap16uses3,152. Zero unresolved in both. At cap16,260later updates and265lift deferrals are retained. Existing adapters remain as controls.

Costs and limits. Four vectors are retained, including cachedG=B·S: at most3d+4live coefficient slots, independent of history length. Slot sums are not peak memory or allocated bytes; charge capacities/metadata and transient workspace. Updates add at mosttwo polynomial inversions and one CRT merge. The current inverse uses ordinary extended Euclid with a conservative O(d M(d))bound; no universal arithmetic/timing advantage is claimed.

Evidence. validation01 pins64transitive inputs and passes with overflow checks. Order, idempotence, ramification, nonunit and CRT controls pass. No native/report failure, cloud/timing trial or fullCargo rerun. Original4,563cohort unchanged. Producer evidence unsigned; full proof/replay in DERIVATION.md.

Sources and next step. Exact persistent predecessor and official FLINT CRT/modular conventions (flintlib.org/doc/nmod_poly/index.html). KB unavailable; no novelty claim. Next evaluate selected-root modes directly modulo S, reserving pair algebra for the both-root component.